

ED211: Engineering Mathematics III

Course Placement a core course for the second year BTech EVD students

Course Format three Lecture hours and two Lab hours per week

Course Content The course ED211 introduces the students to the various mathematical and engineering techniques related to time and frequency analysis and synthesis, Fourier Transformation, Fourier Series, etc. The course demonstrates application of elegant mathematical principles to practical engineering design concepts underlying many algorithms implemented in today's electronics and VLSI systems. The students gain hands-on skills on these topics by performing software laboratory experiments throughout the semester. The specific topics covered are continuous-time Fourier Transform, Fourier Series, effect of sampling, discrete time Fourier Transform, DFT, Z transform, convolution and linear time/shift-invariant systems, design of digital filters, eigenanalysis and its application to OFDM signaling, etc.

Course Description:

- *Introduction to Signals and Systems.* Fourier Transformation, Fourier Series,
- *Discrete Time Signals.* Sampling and its effects
- *Analysis and Processing of Discrete Time Signals.* Discrete Time Fourier Transform (DTFT), Linear Shift Invariant (LSI) systems, impulse response, frequency response
- *Z transformation.* Z transformation and its relation to Laplace transform and DTFT, Poles and Zeros, Region of Convergence.
- *Digital Systems.* Design and analysis of digital filters.

References:

Textbook Oppenheim, A. V., and Schaffer, R. W. (2010). "Discrete-time signal processing," 3rd ed., Prentice Hall.

Reference Books The following books are used as the reference:

1. Kreyszig, E. (2011). "Advanced engineering mathematics," 10th ed., Wiley.
2. Proakis, J. G., and Manolakis, D. G. (2007). "Digital signal processing: Principles, algorithms, and applications," 4th ed., Pearson Prentice Hall.

Grading Policy The assessment method will be as follows:¹

- Two midterm exams: 27.5% each
- Final exam: 30%
- Tutorials, In-class Quizzes and Evaluations: 15%
- Attendance: will follow Institute's attendance policy

¹This is tentative and subject to change.

Course Outcomes this course introduces the students to the engineering knowledge (P1) of the mathematics and engineering fundamentals related to signal analysis and signal processing by linear systems. The students engage in the analysis (P2) and design (P3) of the basic modules of a linear discrete-time and continuous-time system. As part of the Lab/Tutorials, the students conduct investigations of the complex design and analysis problems (P4) and make use the modern tools (P5) such as Matlab and Python. Through the Lab/Tutorial that is team-based and require a formal presentation by the students, they acquire the team-work (P9) and the technical communication (P10) skills.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
✓	✓	✓	✓	✓				✓	✓		